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ML Mini Project Research Paper Summary

TITLE

Used Car Price Prediction using Machine Learning: A Case Study

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RELEVANCE TO OUR PROJECT

This paper directly mirrors our project — both involve predicting used car resale prices using supervised ML regression techniques, log transformation of the target variable, and comparing multiple models including ensemble methods.

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BACKGROUND & MOTIVATION

The used car market has grown significantly in recent years, driven by economic factors that make second-hand vehicles a preferred choice for a large segment of buyers. Rather than committing to bank loans for new vehicles, many customers opt for used cars at more affordable prices. This creates a highly active resale market supported by online e-commerce platforms that connect buyers and sellers directly.

However, a core challenge exists on both sides of this market — neither the buyer nor the seller has a reliable, data-driven way to estimate the fair resale value of a car before making a transaction decision. Pricing is often subjective, inconsistent, and influenced by individual bias rather than actual market patterns. This information asymmetry leads to mispricing, lost revenue for sellers, and overpayment for buyers.

The authors identified this as a machine learning problem. Given that used car prices are determined by a combination of measurable features — such as mileage, fuel type, production year, fiscal power, car brand, and model — a supervised regression model trained on real market data can learn these pricing patterns and produce objective price estimates.

This motivation is directly relevant to our project. The CarDekho dataset we use operates in the same problem space — an online used car marketplace where listed prices are influenced by features like km (mileage), myear (production year), fuel, transmission, oem (brand), and Max Power Delivered. Both the paper and our project treat this as a regression problem with a continuous price target, and both apply log transformation to handle the right-skewed price distribution.

AIM

To build and evaluate a machine learning-based regression system capable of accurately predicting the resale price of used cars, providing an objective pricing tool for buyers and sellers in the second-hand car market.

OBJECTIVES

The paper pursued the following specific objectives:

Objective No.	Objective	Description
1	Collect Real-World Used Car Data	Collect real-world used car data from the online marketplace Avito using web scraping techniques.
2	Data Preprocessing	Preprocess the raw dataset by handling missing values, removing outliers, encoding categorical variables, and scaling numerical features.
3	Exploratory Data Analysis (EDA)	Perform Exploratory Data Analysis (EDA) to understand feature distributions, correlations, and relationships with the target variable (price).
4	Feature Selection using RFE	Apply Recursive Feature Elimination (RFE) to identify the most relevant features influencing car prices.
5	Implement Regression Models	Implement and compare five supervised machine learning regression models: Multiple Linear Regression, KNN Regressor, Random Forest Regressor, Gradient Boosting Regressor, and Artificial Neural Network (ANN).
6	Model Evaluation	Evaluate all models using R^2 Score and RMSE (Root Mean Square Error) to determine the best-performing model.
7	Validate Regression Assumptions	Validate regression assumptions such as multicollinearity, linearity, normality, and homoscedasticity before drawing conclusions.
8	Recommend Best Model	Recommend the most suitable model for deployment based on empirical performance results and evaluation metrics.

PROBLEM STATEMENT

Used car resale pricing lacks objectivity. Existing platforms rely on seller-defined prices which are inconsistent and often inaccurate. The paper addresses the question:

"Can supervised machine learning regression models, trained on real used car listing data, reliably predict the resale price of a vehicle given its key specifications?"

DATASET

The dataset used in this study was collected directly from **Avito** (avito.ma), a Moroccan online used car marketplace, using a custom Python web scraper built with the **BeautifulSoup** library. The scraper crawled all active used car listings and extracted the most relevant features for price prediction. (*Source: Section III, Hankar et al., 2022*)

After cleaning, the final dataset contained **8,000 records** with the following key features:

Feature	Description
mileage	Total kilometres driven
production_year	Year of manufacture
fuel_type	Petrol, Diesel, etc.
fiscal_power	Engine power rating (fiscal horsepower)
mark	Car brand / manufacturer
model	Specific car model
price	Resale listing price — target variable

PREPROCESSING STEPS

(Source: Section III, Hankar et al., 2022)

The raw collected data underwent the following cleaning pipeline:

Step	Action Taken
Missing Values	Imputed using column mean (mileage, fiscal power) to avoid information loss
Outlier Removal	Cleaned from the raw dataset before model training
Feature Selection	Recursive Feature Elimination (RFE) used to rank and retain only price-relevant features
Categorical Encoding	Label encoding applied to fuel type, car mark, and model
Feature Scaling	Standard scaling applied before model training and testing
Log Transformation	Natural log applied to the price variable due to right-skewed distribution — resulted in improved R^2 and reduced RMSE (Source: Section VII, Hankar et al., 2022)

While the prices of the car are not normally distributed and highly right skewed, we applied a natural log transformation on the price variable before training the models. The result is an obvious increase of R^2 score and RMSE errors are decayed even more." — Hankar et al., 2022, Section VII

Linkage to Our Project:

The preprocessing pipeline described in this paper — log transformation of the target variable, label encoding of categorical features, standard scaling, and an 80/20 train-test split — closely mirrors the approach adopted in our project on the CarDekho dataset, validating our methodology choices at each step.

Cardeko dataset Link : <https://www.kaggle.com/datasets/sukritchatterjee/used-cars-dataset-cardexho>

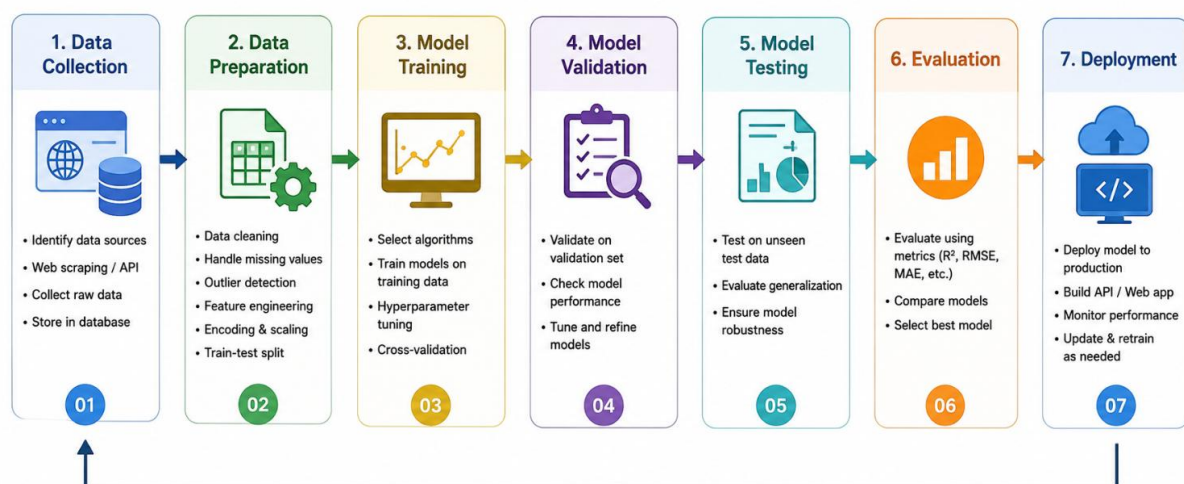
METHODOLOGY

(Source: Section V, Hankar et al., 2022)

The authors framed the problem as a **supervised regression task** — given a set of measurable car features, train a model to predict the continuous target variable (resale price). Five regression models were implemented and compared against each other.

MACHINE LEARNING LIFECYCLE

The overall workflow followed a standard ML lifecycle as illustrated in the paper (Source: Fig. 3, Hankar et al., 2022):



Each stage fed into the next — raw scraped data was cleaned and transformed, models were trained on 80% of the data, validated by tuning hyperparameters, and finally tested on the unseen 20% holdout set.

MODELS IMPLEMENTED

The five regression algorithms tested in the study were:

	Model	Type	Brief Description
1	Multiple Linear Regression (MLR)	Baseline	Fits a linear relationship between multiple input features and price
2	K-Nearest Neighbors Regressor (KNNR)	Instance-based	Predicts price based on the average of K most similar training samples
3	Random Forest Regressor (RFR)	Ensemble — Bagging	Builds multiple decision trees independently and averages their predictions
4	Gradient Boosting Regressor (GBR)	Ensemble — Boosting	Builds trees sequentially, each correcting errors of the previous one
5	Artificial Neural Network (ANN)	Deep Learning	Multi-layer perceptron that learns non-linear feature relationships

FINE-TUNED PARAMETERS

(Source: Table I, Hankar et al., 2022)

All hyperparameters were selected using **Grid Search optimization** to find the combination yielding the highest R^2 and lowest RMSE:

Model	Key Parameters
MLR (Multiple Linear Regression)	production_year_coeff = 0.395 , mileage_coeff = -0.004 , fuel_type_coeff = -0.213 , intercept = 96608.59
KNNR (K-Nearest Neighbors Regressor)	k = 10 , weights = 'uniform'
RFR (Random Forest Regressor)	n_estimators = 100 , criterion = 'mse' , max_depth = 50
GBR (Gradient Boosting Regressor)	loss = 'ls' , max_depth = 6
ANN (Artificial Neural Network)	neurons_per_layer = (32, 16, 8) , activation = 'relu' , loss = 'mse' , optimizer = 'adam'

EVALUATION METRICS

Two metrics were used to evaluate all models (*Source: Section V, Hankar et al., 2022*):

R² Score (Coefficient of Determination) — measures the proportion of variance in car prices explained by the model. Range is 0 to 1; higher is better.

RMSE (Root Mean Squared Error) — measures the average magnitude of prediction error in the same unit as the target variable. Lower is better.

These are the same two primary metrics used in our project, making the results of this paper directly comparable to our own model evaluations.

RESULTS & DISCUSSION

(*Source: Section VII, Hankar et al., 2022*)

After training and evaluating all five models on the 20% holdout test set, the following results were obtained:

MODEL PERFORMANCE COMPARISON (*Source: Table II, Hankar et al., 2022*)

Model	R ² Score	RMSE	Rank
Gradient Boosting Regressor (GBR)	0.80	44,516.20	1st
Random Forest Regressor (RFR)	0.74	44,939.79	2nd
K-Nearest Neighbors Regressor (KNNR)	0.70	51,224.96	3rd
Artificial Neural Network (ANN)	0.67	54,957.98	4th
Multiple Linear Regression (MLR)	0.57	63,933.52	5th

KEY FINDINGS

1. Gradient Boosting Regressor was the clear winner. GBR achieved the highest R^2 of 0.80, meaning it explained 80% of the variance in used car prices. It also recorded the lowest RMSE of 44,516.20, confirming it as the most accurate and reliable model among all tested approaches. (Source: Section VII, Hankar et al., 2022)

2. Ensemble methods outperformed all other approaches. Both GBR and RFR — which are ensemble techniques — ranked first and second respectively. This confirms that ensemble methods capture complex non-linear pricing patterns that simpler models like MLR cannot. The gap between GBR ($R^2=0.80$) and MLR ($R^2=0.57$) is substantial — a 23-point improvement simply by switching from a linear baseline to a boosting ensemble.

3. Linear Regression performed worst. MLR's R^2 of 0.57 suggests that a purely linear relationship between features and price is insufficient. Used car pricing is inherently non-linear — for example, the relationship between mileage and price is not constant across all price ranges.

4. Log transformation was critical. The authors noted that applying log transformation to the price variable before training produced a clear improvement in both R^2 and RMSE across all models. Without it, the highly right-skewed price distribution would have violated regression assumptions and degraded model performance. (Source: Section VII, Hankar et al., 2022)

5. The authors acknowledged limitations. An R^2 of 0.80 for the best model, while the highest among tested approaches, was described by the authors themselves as not a great score. They attributed this to the limited feature set (only 6 features) and suggested that scaling training data and adding more variables could further improve performance. (Source: Section VIII, Hankar et al., 2022)

REGRESSION ASSUMPTIONS VALIDATION (Source: Section VI & VII, Hankar et al., 2022)

One of the most distinctive and rigorous aspects of this paper — rarely seen in standard ML studies — is the explicit validation of classical regression assumptions before drawing conclusions from the best model (GBR). The authors argue that model results carry no statistical significance unless these foundational conditions are verified. (Source: Section VI, Hankar et al., 2022)

Four assumptions were tested:

Assumption	Test Used	Result
No Multicollinearity	Correlation matrix	Satisfied — max correlation only -0.13 between mileage and production year
Linearity	Scatter plots	Satisfied — mileage, fiscal power, production year showed linear trends with price
Normality of Residuals	Residual distribution plot	Satisfied — log transformation brought residuals to near-normal distribution
Homoscedasticity	Residuals vs fitted values	Satisfied — error variance remained constant across all prediction ranges

CONCLUSION OF THE PAPER

The study successfully demonstrated that machine learning regression models can reliably predict used car resale prices. Among all tested models, **Gradient Boosting Regressor emerged as the best performer** with $R^2=0.80$ and the lowest RMSE, outperforming linear regression, KNN, Random Forest, and ANN. The authors concluded that ensemble boosting methods are best suited for this problem and recommended expanding the feature set and scaling training data as future improvements.

CRITICAL ANALYSIS

Aspect	Observation
Strength	Rigorous assumption validation — rarely done in ML papers
Strength	Real-world data collected via web scraping — not a toy dataset
Strength	Log transformation applied correctly — methodologically sound
Limitation	Only 8,000 records and 6 features — small and narrow dataset
Limitation	No XGBoost or AdaBoost tested — stronger boosting variants were missed
Limitation	$R^2=0.80$ acknowledged as insufficient for production deployment

LINKAGE TO OUR PROJECT

The connection between this paper and our project runs deeper than just the shared topic of used car price prediction. At every stage of the ML pipeline, the decisions made in this paper align with our approach.

The most significant parallel is the **treatment of the target variable**. The authors identified that car prices follow a highly right-skewed distribution and applied natural log transformation before training — exactly the same decision we made during our EDA on the CarDekho dataset, where raw `listed_price` had a skewness of 157.12, which reduced to 0.18 after applying `np.log1p()`. The **preprocessing philosophy** is also identical — label encoding for categorical features, standard scaling, and an 80/20 train-test split — mirrored in our project across 41 features.

On the modelling side, the paper's central finding that **Gradient Boosting outperforms all regression baselines** directly supports the structure of our project, where ensemble models are expected to outperform regression baselines — providing literature-backed justification for this comparison.

Our project extends this paper in three ways — a significantly larger dataset (37,038 vs 8,000 records), richer features (41 vs 6), and the inclusion of XGBoost and AdaBoost which this paper did not test. Where they achieved $R^2=0.80$ with 6 features, our richer dataset gives a strong basis to improve on that benchmark.